

Microbiology Galaxy Lab (MGL): The first community-driven gateway for reproducible and FAIR analysis of microbiological data

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Introduction
Microbial research increasingly relies on high-throughput and multi-omics approaches, generating complex datasets. However, identifying and navigating relevant tools and workflows within platforms like Galaxy remains challenging

How do scientists analyse microbial data?

130 responses of the survey conducted among the microbiology research community (March - September 2023) show:

What is your main target?

Bacteria	44.35%
Eukaryotes	22.58%
Mixed Communities	19.35%
Other	7.26%
Virus	6.45%

Which techniques do you use?

Single Organism Genomics	23.89%
Metagenomics	23.08%
(Meta)Transcriptomics	22.27%
Metabarcoding	14.98%
(Meta)Proteomics	8.91%
Metabolomics	3.64%
Other	3.24%

What is your main target?

Functional Analysis	12.68%
Gene Identification	12.36%
Assembly	10.95%
Comparative Analyses	10.17%
Phylogenetic Tree Construction	9.39%
Taxonomic Profiling	9.08%
Single Nucleotide Polymorphism (SNP) Identification	7.87%
Antimicrobial Resistance Detection	7.36%
Pathogen Tracking	5.79%
Metagenome Assembled Genome (MAG) Building	4.85%
Combination Of Multiomic Approaches	5.32%
Predictive Metagenomics	3.44%
Other	0.94%

Which analysis do you use or would like to do?

Yes	77.69%
No	11.54%
Unknown	10.77%

Number of Responses

Scientists study different organisms using different techniques and analyses and want to use Galaxy

Could scientists use Galaxy for microbial data analysis?

Resources for microbial analysis in Galaxy are highly used for a wide range of key computational functions

Cumulative number of tool suites, tutorials and workflows

Suite users (last 5 years) on main servers

Suite runs (last 5 years) on main servers

Do scientists use Galaxy for publications?

28% of publications citing Galaxy are microbiology-related papers

Number of publications citing Galaxy papers

Topics

- Ecosystem Dynamics & Biodiversity
- Health & Disease
- Non microbiology-related
- Ecosystem Dynamics & Biodiversity, Health & Disease
- Unlabeled microbiology-related

How does MGL help scientists find the right resources for microbial data analysis?

A

Tools sorted by type of analysis and/or tool suites

Direct links to resources (tools, workflows, training material) for **Microbial Isolates** data analysis

Direct links to resources (tools, workflows, training material) for **Microbiome** data analysis

Links to curated tools grouped by EDAM operations

B

History of analysis and files

Links to curated workflows grouped by FAIRness levels

Links to support and help

C

Links to the community

Contributors

Microbiology Galaxy Lab (MGL) interface (A) in comparison with default Galaxy interfaces on the UseGalaxy server (B, C)

MGL availability:

microbiology.usegalaxy.org, .eu, .org.au, .fr (and more Galaxy servers to come)

For more information about the MGL and our roadmap, check our preprint

bioRxiv

THE PREPRINT SERVER FOR BIOLOGY

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